Exercise J02

Q1. Find the general values of q for which the quadratic function $(\sin\theta)\, x^2 + (2\cos\theta)x + \frac{\cos\theta + \sin\theta}{2}$ is the square of a linear function.

Ans. $2n\pi + \frac{\pi}{4} or (2n+1)\pi - \tan^{-1} 2, n \in I$

Sol. -

Q2. Solve the equality: $2 \sin 11x + \cos 3x + \sqrt{3} \sin 3x = 0$

Ans. x$= \frac{n\pi}{7} - \frac{\pi}{84}$ or x $= \frac{n\pi}{4} + \frac{\pi}{48}, n \in 1$

Sol. -

Q3. Find all the values of θ satisfying the equation; $\sin \theta + \sin 5\theta = \sin 3\theta$ such that 0 $0 \le \theta \le \pi$.

Ans. $0, \frac{\pi}{6}, \frac{\pi}{3} \frac{2\pi}{3}, \frac{5\pi}{6}, \&\pi$

Sol. -

Q4. Find all values of θ between 0° & 180° satisfying the equation;$\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0$

Ans. 30° , 45° , 90° , 135° , 150°

Sol. -

Q5. The positive integer value of n > 3 satisfying the equation $\frac{1}{sin\left(\frac{\pi}{n}\right)} = \frac{1}{sin\left(\frac{2\pi}{n}\right)} + \frac{1}{sin\left(\frac{3\pi}{n}\right)}$ is

Ans. 7

Sol. Given $\frac{1}{\sin\frac{\pi}{n}} = \frac{1}{\sin\frac{2\pi}{n}} + \frac{1}{\sin\frac{3\pi}{n}} \& n > 3$

$\Rightarrow \frac{1}{\sin\frac{\pi}{n}} - \frac{1}{\sin\frac{3\pi}{n}} = \frac{1}{\sin\frac{2\pi}{n}}$

$\Rightarrow \frac{\sin\frac{3\pi}{n} - \sin\frac{\pi}{n}}{\sin\frac{\pi}{n}\sin\frac{3\pi}{n}} = \frac{1}{\sin\frac{2\pi}{n}}$

$\Rightarrow \frac{2\cos\frac{2\pi}{n}\sin\frac{\pi}{n}}{\sin\frac{\pi}{n}\sin\frac{3\pi}{n}} = \frac{1}{\sin\frac{2\pi}{n}}$

$\Rightarrow 2\cos\frac{2\pi}{n}\sin\frac{2\pi}{n} = \sin\frac{3\pi}{n}$

$\Rightarrow \sin\frac{4\pi}{n} = \sin\frac{3\pi}{n} \Rightarrow \frac{4\pi}{n} = K\pi + (-1)^K \frac{3\pi}{n}$

If K = 2m then $\frac{\pi}{n} = 2m\pi$

$\Rightarrow n = \frac{1}{2m}$ $\Rightarrow n = \frac{1}{2}, \frac{1}{4}, \frac{1}{6}$ ....

If k = 2m + 1 then $\frac{7\pi}{n} = (2m+1)\pi$

$\Rightarrow n = \frac{7}{2m+1}$ $\Rightarrow n = 7, \frac{7}{3}, \frac{7}{5}$.......

Given n > 3 ⇒ n = 7

Q6. Find all value of θ, between 0 & π, which satisfy the equation; cos θ . cos 2 θ . cos 3 θ = 1/4.

Ans. $\frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$

Sol. -

Q7. Find the general solution of the equation , $2 + \tan x \cdot \cot \frac{x}{2} + \cot x \cdot \tan \frac{x}{2} = 0$

Ans. $x = 2n\pi \pm \frac{2\pi}{3}, n \in 1$

Sol. -

Q8. Sum of all the solutions in [0, 4π] of the equation tan x + cot x + 1 = cos $\left(x + \frac{\pi}{4}\right)$ is k π then the value of k is

Ans. 3.5

Sol. -

Q9. Find the range of y such that the equation , y + cos x = sin x has a real solution. For y = 1 , find x such that 0 < x < 2π.

Ans. $-\sqrt{2} \le y \le \sqrt{2}; \frac{\pi}{2}, \pi$

Sol. -

Q10. Let f (x) = $\sin^6 x + \cos^6 x + k(\sin^4 x + \cos^4 x)$ for some real number k. Determine

(a) all real numbers k for which f (x) is constant for all values of x.

(b) all real numbers k for which there exists a real number 'c' such that f (c) = 0.

(c) If k = – 0.7, determine all solutions to the equation f (x) = 0.

Ans. (a) $-\frac{3}{2}$; (b) k $\in \left[-1, -\frac{1}{2}\right]$; (c) x = $\frac{n\pi}{2} \pm \frac{\pi}{6}$

Sol. -

Q11. Solve the equation for x, $5^{\frac{1}{2}} + 5^{\frac{1}{2}\log^5(sinx)} = 15^{\frac{1}{2}+\log_{15}\cos x}$

Ans. x = $2n\pi + \frac{\pi}{6}, n \in I$

Sol. -

Q12. Solve for x , the equation $\sqrt{13-18tanx} = 6\tan x - 3$, where $-2\pi < x < 2\pi$.

Ans. $\alpha - 2\pi$; $\alpha - \pi, \alpha, \alpha + \pi$, where $\tan\alpha = \frac{2}{3}$

Sol. -

Q13. Determine the smallest positive value of x which satisfy the equation,
$\sqrt{1+\sin 2x} - \sqrt{2}\cos 3x = 0$.

Ans. $x = \pi/16$

Sol. -

Q14. $2sin\left(3x+\frac{\pi}{4}\right) = \sqrt{1+8\sin 2x . cos^2 2x}$

Ans. $x = 2n\pi + \frac{\pi}{12}$ or $2n\pi + \frac{17\pi}{12}; n \in I$

Sol. -

Q15. Find the general solution of the trigonometric equation
$3^{\left(\frac{1}{2}+\log(\cos x+\sin x)\right)} - 2^{\log_2(\cos x - \sin x)} = \sqrt{2}.$

Ans. $x = 2np + \frac{\pi}{12}$

Sol. -

Q16. If sum of all the solutions of the equation $\log_{\frac{9x-x^2-14}{7}}(\sin 3x - \sin x) = \log_{\frac{9x-x^2-14}{7}}$ $\cos 2x$ is equal to $\frac{k\pi}{6}$, then find the value of k.

Ans. 18

Sol. -

Q17. If the quadratic equation $x^2 + (2 - \tan\theta)x - (1 + \tan\theta) = 0$ has two integral roots, then sum of all possible values of θ in interval $(0, 2\pi)$ is kp. Find the value of k.

Ans. 4

Sol. -

Q18. Solve the inequality $\sin 2x > \sqrt{2}\sin^2 x + (2-\sqrt{2})\cos 2x$.

Ans. $n\pi + \frac{\pi}{8} < x < n\pi + \frac{\pi}{4}$

Sol. -

Q19. For $0 < x, y < \pi$, find the number of ordered pairs (x, y) satisfying the system of equations

$\cot^2(x-y) - \left(1+\sqrt{3}\right)\cot(x-y) + \sqrt{3} = 0$ and $\cos y = \frac{\sqrt{3}}{2}$.

Ans. 2

Sol. -

Q20. If the number of solutions satisfying the equation

$$(\sin\theta - 1)(2\sin\theta - 1)(3\sin\theta - 1).......(n\sin\theta - 1) = 0$$

(where $n \in N$) in $[0, \pi]$ is 9, then find the number of ordered pairs (x, y) satisfying the equation

$3 + \operatorname{cosec}^2 x + 2^{\sin^2 y} = n$, where $0 \le x, y \le 4\pi$.

Ans. 20

Sol. -

Q21. Solve x and y : $4^{\sin x}, + 3^{\frac{1}{\cos y}} = 11,\quad 5.16^{\sin x} - 2.3^{\frac{1}{\cos y}} = 2$

Ans. $x = n\pi + (-1)^n \frac{\pi}{6}$ and $y = 2m\pi \pm \frac{\pi}{3}$, where $m, n \in I$

Sol. -

Q22. The number of values of θ in the interval $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ such that $\theta \neq \frac{n\pi}{5}$ for $n = 0, \pm 1, \pm 2$ and $\tan\theta = \cot 5\theta$ as well as $\sin 2\theta = \cos 4\theta$, is

Ans. 3

Sol. Given $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \& \theta \neq \frac{n\pi}{5}$ for $n = 0, \pm 1, \pm 2$

$\sin 2\theta = \cos 4\theta$

$\sin 2\theta = 1 - 2\sin^2 2\theta$

$2\sin^2 2\theta + \sin 2\theta - 1 = 0$

$\Rightarrow \sin 2\theta = -1 \ or \ \sin 2\theta = \frac{1}{2}$

$\therefore \theta = -\frac{\pi}{4}, \frac{\pi}{12}, \frac{5\pi}{12}$

Now $\tan\theta = \cot 5\theta$ (Given)

All three obtained values of θ satisy the given equation.

Hence, number of values of θ are 3.

Paragraph

Let $f(x) = \sin^3 x + 8\cos^6 x - 3$, $g(x) = 2 + 3\sin^2 x \cos x$ and $h(x) = \sin x + 3\cos 2x \sin x$.

On the basis of above information, answer the following questions :

Q23. If $g(x) + h(x) = 0$, then number of solutions in the interval $x \in [0, 2\pi]$ is-

(A) 1 (B) 3 (C) 5 (D) 6

Ans. C

Sol. g(x) + h(x) = 0

$\Rightarrow 2 + 3(\sin 2x \cos x + \cos 2x \sin x) + \sin x = 0$

$\Rightarrow 2 + 3 \sin 3x + \sin x = 0$

$\Rightarrow 9 \sin x - 12 \sin^3 x + \sin x + 2 = 0$

$\Rightarrow 12 \sin^3 x - 10 \sin x - 2 = 0$

$\Rightarrow 6\sin^3 x - 5\sin x - 1 = 0$

$\Rightarrow (\sin x - 1)(6 \sin^2 x + 6 \sin x + 1) = 0$

$\Rightarrow \sin x = 1 \quad \text{or} \quad \sin x = \dfrac{-3-\sqrt{3}}{6} \quad \text{or} \quad \sin x = \dfrac{-3+\sqrt{3}}{6}$

In $0 \le x \le 2\pi$

$\sin x = 1$ → only one solution

$\sin x = \dfrac{-3-\sqrt{3}}{6}$ → two solutions

$\sin x = \dfrac{-3+\sqrt{3}}{6}$ → two solutions

$\therefore$ No. of solution is 5.

Paragraph

Let $f(x) = \sin^3 x + 8\cos^6 x - 3$, $g(x) = 2 + 3\sin^2 x \cos x$ and $h(x) = \sin x + 3\cos 2x \sin x$.

On the basis of above information, answer the following questions :

Q24. If f(x) + g(x) = 0, then number of solutions in the interval $x \in \left[-\dfrac{\pi}{2}, \dfrac{5\pi}{2}\right]$ is –

(A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. $f(x) + g(x) = 0 \quad ; \quad x \in \left[-\dfrac{\pi}{2}, \dfrac{5\pi}{2}\right]$

$\Rightarrow \sin^3 x + 8\cos^6 x - 3 + 2 + 3 \sin 2x \cos x = 0$

$\Rightarrow (\sin x)^3 + (2\cos^2 x)^3 + (-1)^3 = 3 \sin x (2\cos^2 x)(-1)$

$\Rightarrow \sin x + 2\cos^2 x - 1 = 0$ or

$\underbrace{\sin x = -1 = 2\cos^2 x}_{\text{Notpossible}} \qquad \left\{\begin{matrix} \because a^3 + b^3 + c^3 = 3abc \\ \text{if} a = b = c \text{or} a + b + c = 0 \end{matrix}\right\}$

$\Rightarrow \sin x + 2(1 - \sin^2 x) - 1 = 0$

$\Rightarrow 2\sin^2 x - \sin x - 1 = 0$

$\Rightarrow (\sin x - 1)(2 \sin x + 1) = 0$

$\Rightarrow \sin x = 1 \text{ or } \sin x = \dfrac{-1}{2}$

In $\left[-\dfrac{\pi}{2}, \dfrac{5\pi}{2}\right]$

Number of solutions is 5.

Q25. If $\left(\sin 2x+\sqrt{3}\cos 2x\right)^2-5=\cos\left(2x-\frac{\pi}{6}\right)$ then the value of cos x can be

(A) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (B) $\frac{\sqrt{6}-\sqrt{2}}{4}$ (C) $-\left(\frac{\sqrt{6}+\sqrt{2}}{4}\right)$ (D) $\frac{\sqrt{2}-\sqrt{6}}{4}$

Ans. B, D

Sol. $$\left(\sin 2x+\sqrt{3}\cos 2x\right)^2=5+\cos\left(2x-\frac{\pi}{6}\right)$$

$$\Rightarrow 4\left(\frac{\sin 2x}{2}+\frac{\sqrt{3}}{2}\cos 2x\right)^2=5+\cos\left(2x-\frac{\pi}{6}\right)$$

$$\Rightarrow 4\cos^2\left(2x-\frac{\pi}{6}\right)-\cos\left(2x-\frac{\pi}{6}\right)=5$$

$$\Rightarrow 4\cos^2\left(2x-\frac{\pi}{6}\right)-\cos\left(2x-\frac{\pi}{6}\right)-5=0$$

$$\Rightarrow \left(4\cos\left(2x-\frac{\pi}{6}\right)-5\right)\left(\cos\left(2x-\frac{\pi}{6}\right)+1\right)=0$$

$$\Rightarrow \underbrace{\cos\left(2x--\frac{\pi}{6}\right)=\frac{5}{4}}_{\text{Notpossibleas}\frac{5}{4}>1.} \quad \text{or} \quad \cos\left(2x-\frac{\pi}{6}\right)=-1$$

$$\Rightarrow \cos\left(2x-\frac{\pi}{6}\right)=-1$$

$$\Rightarrow 2x-\frac{\pi}{6}=(2n-1)\pi$$

$$\Rightarrow x=(2n-1)\frac{\pi}{2}+\frac{\pi}{12}$$

$$\Rightarrow x=n\pi-\frac{5\pi}{12};n\in I$$

$$\Rightarrow x=-\frac{5\pi}{12},\frac{7\pi}{12},\frac{-17\pi}{12},........$$

$$\cos\left(\frac{7\pi}{12}\right)=\cos\left(\pi-\frac{5\pi}{12}\right)=-\cos\frac{5\pi}{12}=\frac{\sqrt{2}-\sqrt{6}}{4}$$

$$\cos\left(-\pi-\frac{5\pi}{12}\right)=\cos\left(\pi+\frac{5\pi}{12}\right)=-\cos\left(\frac{5\pi}{12}\right)=\frac{\sqrt{6}-\sqrt{2}}{4}$$

Q26. If $|cosx|^{sin^2x-\frac{3}{2}sinx+\frac{1}{2}}=1$. Then possible values of x are :

(A) $n\pi$ or $n\pi+(-1)^n\frac{\pi}{6}, n\in I$ (B) $n\pi$ or $2n\pi+\frac{\pi}{2}$ or $n\pi+(-1)^n\frac{\pi}{6}, n\in I$

(C) $n\pi + (-1)^n \dfrac{\pi}{6}, n \in I$ (D) $n\pi, n \in I$

Ans. A, C, D

Sol. $|\cos x|^{\sin^2 x - \frac{3}{2}\sin x + \frac{1}{2}} = 1$

Case I :- $\sin^2 x - \dfrac{3}{2}\sin x + \dfrac{1}{2} = 0$

$\Rightarrow 2\sin^2 x - 3\sin x + 1 = 0$

$\Rightarrow (2\sin x - 1)(\sin x - 1) = 0$

$\Rightarrow \sin x = \dfrac{1}{2}$ or $\sin x = 1$

$\Rightarrow x = n\pi + (-1)^n \dfrac{\pi}{6}$ *or* $x = (4n+1)\dfrac{\pi}{2}; n \in I$

Case II :- $|\cos x| = 1$

$\Rightarrow x = n\pi$

Q27. If $\dfrac{\sin^2 2x + 4\sin^4 x - 4\sin^2 x\cos^2 x}{4 - \sin^2 2x - 4\sin x} = \dfrac{1}{9}$ and $0 < x < \pi$, then value of x is :

(A) $\dfrac{\pi}{3}$ (B) $\dfrac{\pi}{6}$ (C) $\dfrac{2\pi}{3}$ (D) $\dfrac{5\pi}{6}$

Ans. B, D

Sol. -

Q28. $5\sin x - 12\cos x = -13\sin 3x$, if $\phi = \sin^{-1}\left(\dfrac{12}{13}\right)$, then :

(A) $\sin\left(2x - \dfrac{\phi}{2}\right) = 0$ (B) $\sin\left(x + \dfrac{\phi}{2}\right) = 0$ (C) $\cos\left(x + \dfrac{\phi}{2}\right) = 0$

(D) $\cos\left(2x + \dfrac{\phi}{2}\right) = 0$

Ans. A, C

Sol. -

Q29. If $\sin 5\theta = a\sin^5\theta + b\sin^3\theta + c\sin\theta + d$, $\forall\ \theta \in R$, then :

(A) $a + b + c + d = 1$ (B) $a + b + c = 1$ (C) $5a + 4b = 0$ (D) $b + 4c = 0$

Ans. A, B, C, D

Sol. Given $\sin 5\theta = a\sin^5\theta + b\sin^3\theta + c\sin\theta + d \quad \forall\theta\in R$

$\Rightarrow \sin 5\theta = \sin(3\theta + 2\theta)$

$= \sin 3\theta\cos 2\theta + \cos 3\theta\sin 2\theta$

$= 2\sin\theta\cos\theta(4\cos^3\theta - 3\cos\theta) + (1 - 2\sin^2\theta)(3\sin\theta - 4\sin^3\theta)$

$= 8\sin\theta\cos^4\theta - 6\sin\theta\cos^2\theta + 3\sin\theta - 4\sin^3\theta - 6\sin^3\theta + 8\sin^5\theta$

$= 8 \sin \theta + 8 \sin^5\theta - 16 \sin^3\theta - 3 \sin \theta - 4 \sin^3\theta + 8 \sin^5\theta$

$= 16 \sin^5\theta - 20 \sin^3\theta + 5 \sin \theta$

$\Rightarrow a = 16, b = -20, c = 5, d = 0$

$\Rightarrow a + b + c + d = 1$

$a + b + c = 1$

$5a + 4b = 80 - 80 = 0$

$\& \ b + 4c = 0$

Q30. Match The column

Column I	Column II
(A) Solution of $\lvert\tan \theta\rvert + \lvert\cot \theta\rvert \leq 2$ is	(P) $2 n \pi + \frac{5\pi}{4}$, $n \in I$
(B) Solution of $(\sqrt{2} \cos \theta + 1)^2 + (\tan \theta - 1)^2 = 0$ is	(Q) Null set
(C) Solution of $\sin\left(\frac{11}{10}\cos\theta\right) = \cos\left(\frac{11}{10}\sin\theta\right)$ is	(R) $(2 n + 1) \frac{\pi}{4}$, $n \in I$
(D) Solution of $\sin x = 2^{x-1} + 2^{1-x}$ is	(S) $n p + (-1)^n \frac{\pi}{6}$, $n \in I$

(A) (A) R, (B) P, (C) Q, (D) Q (B) (A) P, (B) R, (C) Q, (D) Q

(C) (A) Q, (B) P, (C) R, (D) Q (D) (A) R, (B) Q, (C) Q, (D) P

Ans. 5ef6f5f7239ecc768b1932ce

Sol. -

Q31.

Column I (Equation)	Column II (General Solution)
(A) $\sin x + \cos x = 1$	(P) $2 n \pi$, $n \in I$
(B) $\sin x + \cos x = -1$	(Q) $(2 n + 1) \pi$, $n \in I$
(C) $\lvert\sin x\rvert + \lvert\cos x\rvert = 1$	(R) $(4 n + 1) \frac{\pi}{2}$, $n \in I$
(D) $\lvert\sin x + \cos x\rvert = 1$	(S) $(4 n - 1) \frac{\pi}{2}$, $n \in I$

(A) (A)QS, (B) P, (C) PRS, (D) PQRS (B) (A)PRQR, (B) Q, (C) PS, (D) PQRS

(C) (A)PR, (B) QS, (C) PQRS, (D) PQRS (D) (A)PR, (B) PQRS, (C) PQRS, (D) PS

Ans. 5ef707339fff15766e3064be

Sol. -

Q32. If α and β are the roots of the equation, $a \cos \theta + b \sin \theta = c$ then match the entries of column-I with the entries of column-II.

Column-I	Column-II
(A) $\sin \alpha + \sin \beta$	(P) $\frac{2b}{a+c}$
(B) $\sin \beta \cdot \sin \beta$	(Q) $\frac{c-a}{c+a}$

(C) $\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}$ (R) $\frac{2bc}{a^2+b^2}$

(D) $\tan\frac{\alpha}{2}.\tan\frac{\beta}{2}=$ (S) $\frac{c^2-a^2}{a^2+b^2}$

(A) (A) S; (B) R; (C) P; (D) Q (B) (A) R; (B) S; (C) P; (D) Q

(C) (A) R; (B) S; (C) Q; (D) P (D) (A) Q; (B) P; (C) S; (D) R

Ans. 5ef706234c298c76832d52a1

Sol. -